AXA CLEANING

Commercial Website, Booking Platform & Administration Portal

Software Requirements Specification (SRS)

Document field	Value
Document version	1.0
Prepared for	AXA Cleaning
Prepared by	Buddika A. Silva
Target hosting	Azure Static Web Apps with GitHub-based CI/CD
Document date	25 October 2026
Status	Development baseline

Purpose
This document is the authoritative functional, technical, security, UX, and delivery specification for an AI-assisted development team building the AXA Cleaning platform.

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Document Control

Version	Date	Owner	Change
1.0	25 Oct 2026	Buddika A. Silva	Initial complete SRS and master developer prompt

Important implementation decision
Do not store or compare plaintext passwords. The front-page “Admin Portal” button must lead to a protected sign-in experience using Microsoft Entra ID or another secure identity provider. Role authorization must be enforced server-side, not merely hidden in the interface.

1. Executive Summary

AXA Cleaning requires a production-ready digital platform consisting of a public commercial website, a customer booking experience, and a secured administration portal. The system will be maintained in GitHub and deployed to Azure Static Web Apps. Server-side operations will run through protected APIs, with persistent data stored in an Azure-managed database and transactional email sent through a supported email service.

Customers will explore services and package options, compare inclusions, select an address and preferred timeslot, submit a booking request, and receive a confirmation email. AXA Cleaning staff will review the request against cleaner availability, then accept, modify, reject, or create bookings. Every booking change will be recorded and communicated. The solution must work equally well on mobile, tablet, and desktop.

Business objective	Success indicator
Increase qualified bookings	Visitors can compare packages and submit a complete request in under five minutes.
Reduce scheduling conflicts	No confirmed assignment can double-book an active cleaning agent.
Improve response quality	Admins see pending requests, availability, status history, and recommended actions in one dashboard.
Protect operations	Identity, role checks, audit logs, input validation, rate limiting, and least privilege are applied.
Enable future growth	Packages, service areas, staff, pricing, and content are configurable without code changes.

2. Vision, Goals and Scope

2.1 Product vision

Create a premium, trustworthy and memorable online presence for AXA Cleaning while connecting marketing, booking, workforce availability, operations, and customer communications in one coherent platform.

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2.2 In scope

- Responsive public marketing website with premium 3D-inspired visuals and performance-conscious animation.
- Service/package catalogue with filters, comparison, inclusions, exclusions, pricing rules, duration, and add-ons.
- Guest customer booking request journey with availability preferences and address/service details.
- Protected admin portal with Super Admin, Manager, and Regular Admin roles.
- Booking workflow: draft, requested, under review, awaiting customer response, accepted, rejected, cancelled, completed, and no-show.
- Cleaning-agent records, skills, shifts, leave, service areas, assignments, and calendar availability.
- Email notifications with a stable booking reference and reply/thread correlation.
- Operational dashboard, search, filtering, audit history, exports, and basic reporting.
- GitHub repository, automated Azure deployment, environment separation, testing, monitoring, and documentation.

2.3 Out of scope for version 1

- Native iOS or Android app. The responsive web app is the supported mobile experience.
- Online card payment and refunds unless separately approved. The design shall preserve an integration point.
- Payroll, full HR, route optimization, GPS tracking, or biometric attendance.
- Customer self-service accounts. Version 1 supports guest booking with secure booking-management links; accounts may be added later.
- Unrestricted customer selection of a specific cleaner. Staff assignment remains an administrative decision.

3. Stakeholders and User Roles

Role	Key capabilities	Restrictions
Public visitor / customer	Browse website, compare packages, request a booking, receive updates, reply to booking email, use secure booking link.	No access to internal schedules, other customers, internal notes, or staff details.
Regular Admin	View dashboard; review, create, accept, modify, reject, cancel, complete bookings; view availability; assign eligible agents; add notes.	Cannot manage Super Admins, system-wide security, secrets, or protected configuration.
Manager	All Regular Admin functions; manage packages, agents, service areas, content, operational users, reports and schedules.	Cannot view, edit, disable, delete, demote, reset, or otherwise act upon a Super Admin account.
Super Admin	Unrestricted business-application access; manage roles, managers, system settings, integrations and audit review.	Cannot bypass platform security controls or erase immutable audit records.
Cleaning Agent	Represented as a schedulable resource in v1; optional restricted portal can be enabled later.	No admin access by default.
System / automation	Validates conflicts, sends email, writes audit events, creates reminders and status transitions.	Uses managed identity/service credentials and least privilege.

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3.1 Authorization matrix

Capability	Regular Admin	Manager	Super Admin
View/operate bookings	Yes	Yes	Yes
View agent availability	Yes	Yes	Yes
Manage agent schedules	Optional by permission	Yes	Yes
Manage packages/content	No	Yes	Yes
Create Regular Admin	No	Yes	Yes
Create Manager	No	Yes	Yes
Manage Super Admin	No	No	Yes
Manage integrations/settings	No	Limited	Yes
View audit logs	Own/operational	Yes	Yes
Export operational data	If granted	Yes	Yes

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4. Solution Architecture

4.1 Recommended architecture

Layer	Recommended service / technology	Responsibility
Source and CI/CD	GitHub repository + GitHub Actions	Version control, pull requests, automated tests, preview environments and deployment.
Web front end	React + TypeScript + Vite on Azure Static Web Apps	Public website, booking interface and admin single-page application.
Authentication	Microsoft Entra ID through Azure Static Web Apps custom authentication	Secure staff sign-in, tenant restriction and identity claims.
Authorization	Application RBAC in API plus Static Web Apps route rules	Server-side enforcement of Super Admin, Manager and Regular Admin permissions.
API	Azure Functions (TypeScript/Node.js) linked to Static Web Apps	Bookings, availability, users, packages, agents, notifications, audit, reports.
Database	Azure SQL Database (preferred) or Azure Cosmos DB if approved	Transactional data, constraints, concurrency and reporting.
Email	Azure Communication Services Email or approved transactional provider	Acknowledgements, decisions, changes, reminders, and reply handling.
Secrets	Azure Key Vault + managed identities	Secrets, connection strings, certificates and rotation.
Observability	Application Insights + Azure Monitor	Telemetry, errors, performance, alerts and operational tracing.
Assets	Azure Blob Storage or optimized repository assets	Package media and controlled content assets.

Architecture rule
A static front end alone is not sufficient. All sensitive operations, booking checks, role enforcement, email dispatch, and data writes must run through authenticated server-side APIs.

4.2 Logical application areas

- Public site: Home, Services, Packages, Compare, About, Areas Served, Reviews, FAQs, Contact, Booking.
- Booking app: package selection, add-ons, property details, preferred date/time, contact details, review, consent, confirmation.
- Admin portal: dashboard, booking board, calendar, availability, customers, agents, packages, content, users, reports, audit, settings.

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- API and jobs: validation, conflict detection, notifications, reminders, status transitions, reports, and inbound email correlation.

5. Functional Requirements

5.1 Public commercial website

ID	Requirement
FR-PUB-001	The home page shall clearly present AXA Cleaning's value proposition, primary services, service area, trust indicators and booking call-to-action.
FR-PUB-002	The site shall provide editable pages/sections for Home, About, Services, Packages, Reviews, FAQ, Contact, Privacy and Terms.
FR-PUB-003	The package catalogue shall support name, short/long description, base price or "from" price, duration, size limits, inclusions, exclusions, recommended frequency, badges, media, availability and display order.
FR-PUB-004	Customers shall compare two to four packages side by side, with differences visually highlighted and a clear selection action.
FR-PUB-005	The interface shall support add-on services with price and duration adjustments.
FR-PUB-006	The public site shall include an Admin Portal link that routes to secure authentication without exposing credentials or administrative content.
FR-PUB-007	The site shall provide search-engine metadata, structured data where appropriate, social sharing metadata, sitemap and robots configuration.
FR-PUB-008	The site shall include contact forms protected by validation, anti-spam controls and rate limiting.

5.2 Customer booking

ID	Requirement
FR-BKG-001	A customer shall select one package and zero or more compatible add-ons.
FR-BKG-002	The customer shall enter service address, property type, approximate size, number of rooms/bathrooms, access notes, special requirements and optional attachments if enabled.
FR-BKG-003	The customer shall provide name, email, phone and consent to privacy/booking terms.
FR-BKG-004	The customer shall select a preferred date and time window plus at least one optional alternative. The interface must label this as a request until accepted.
FR-BKG-005	The system shall calculate an estimated total and estimated duration using the active pricing configuration. Admin users may adjust estimates with a recorded reason.

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ID	Requirement
FR-BKG-006	Before submission, the customer shall review all entered details and edit any step.
FR-BKG-007	On successful submission, the system shall create a unique human-readable reference such as AXA-2026-000123, show a confirmation page and send an acknowledgement email.
FR-BKG-008	Duplicate submissions shall be prevented using idempotency and UI submission locking.
FR-BKG-009	A secure, expiring, revocable link may allow the customer to view booking status, approve a proposed change, request cancellation or add a reply.
FR-BKG-010	The system shall store customer-visible messages separately from internal admin notes.

5.3 Cleaning-agent availability

ID	Requirement
FR-AVL-001	Maintain agent name, status, contact data, skills, service areas, working hours, maximum jobs/day, travel buffer and notes.
FR-AVL-002	Maintain recurring work patterns, one-off shift changes, leave, sickness, training and blocked time.
FR-AVL-003	Availability shall subtract confirmed/provisional assignments, blocked time, travel buffer and leave from working windows.
FR-AVL-004	The dashboard shall display day, week and month views with available, provisional, busy and unavailable states.
FR-AVL-005	The system shall warn and block confirmation when an assignment overlaps another active assignment unless an authorized override is recorded.
FR-AVL-006	Admins shall filter eligible agents by date, duration, skills, service area and status.
FR-AVL-007	Multi-agent bookings shall reserve every assigned agent for the booking duration and buffers.

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6. Booking and Scheduling Rules

6.1 Booking state model

Status	Meaning	Allowed next states
Draft	Created by admin but not submitted/communicated.	Requested, Cancelled
Requested	Customer request received; no commitment yet.	Under Review, Rejected, Cancelled
Under Review	Staff checking scope, pricing and availability.	Awaiting Customer, Accepted, Rejected, Cancelled
Awaiting Customer	AXA proposed a change and needs customer response.	Under Review, Accepted, Rejected, Cancelled
Accepted	Date/time and assignment confirmed.	Modified, In Progress, Cancelled, No-show
Modified	Confirmed booking changed and re-notified.	Accepted, Awaiting Customer, Cancelled
In Progress	Service has started.	Completed, Cancelled
Completed	Service delivered.	Closed
Rejected	Request declined with customer-visible reason.	Closed
Cancelled	Cancelled by customer or AXA, with actor/reason.	Closed
No-show	Access/customer issue prevented service.	Closed
Closed	Final state retained for records.	None

6.2 Conflict and capacity rules

- A request may be received even when capacity is uncertain; it must not be represented as confirmed.
- Acceptance requires at least one eligible active agent, adequate duration, required skills, service-area compatibility, and no overlap including buffers.
- The API must re-check availability within the same transaction that confirms the booking to prevent race conditions.
- A provisional hold may have an expiry time. Expired holds release capacity automatically.
- Admin overrides require an elevated permission, a reason, and an audit event.
- Changes to date, time, duration, package, price, address, or assigned agents create a new booking revision.

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7. Notifications and Email Threading

7.1 Required notifications

Event	Customer email	Internal notification
Booking requested	Acknowledgement, summary, reference, requested slot, “not yet confirmed” statement.	New request alert to configured recipients and in-app notification.
Accepted	Confirmed date/time, service, estimate, address summary, contact route.	Assignment shown on staff dashboard.
Modification proposed	Old vs new details and secure accept/respond action.	Task remains open until customer action or admin resolution.
Rejected	Clear, respectful reason and optional alternatives/contact details.	Status and reason logged.
Cancelled	Actor, reason, effective date and any next steps.	Capacity released and audit event recorded.
Reminder	Configurable reminder before service.	Optional assignment reminder.
Completed	Completion notice and optional feedback request.	Completion metric updated.

7.2 Thread continuity

Every booking shall have a stable conversation/thread identifier. Outbound messages for that booking should preserve a consistent subject containing the booking reference and use provider-supported reply/thread headers, including Message-ID, In-Reply-To and References where applicable. If inbound reply processing is enabled, the reply address shall encode or map to the booking and validated sender. All customer-visible correspondence shall also be stored in the booking timeline.

- Company sender name and address must be configurable, for example AXA Cleaning <bookings@company-domain>.
- Template content, logo, footer, company contact details and legal text must be centrally configurable.
- A provider message ID, delivery status, template version, timestamp and failure reason must be logged.
- The system must retry transient failures with bounded exponential backoff and place persistent failures in an operational queue.
- Email content must not expose internal notes, staff private data, secrets, or full authentication links in logs.

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8. Administration Portal

8.1 Dashboard

- KPI cards: new requests, awaiting action, accepted today, jobs this week, unassigned bookings, available agents, cancellations, and overdue tasks.
- Availability panel: timeline/calendar of active agents with filters and color-coded states.
- Action queue: oldest pending requests, proposed changes awaiting customer response, email failures, conflicts and expiring holds.
- Charts: booking volume, acceptance rate, package mix, utilization and cancellation reasons.
- Global search by booking reference, customer, phone, email, address and agent.
- Dashboard filters persist in the user session and respect the user's authorization scope.

8.2 Booking workspace

- List, kanban and calendar views with saved filters.
- Booking detail drawer/page with summary, customer data, package, add-ons, status, revisions, assignment, price, messages, internal notes, audit trail and email delivery events.
- Actions: create, accept, propose modification, assign/reassign, reject, cancel, mark in progress/completed/no-show, resend message, copy secure link, export or print summary.
- All material updates require validation; destructive or customer-impacting actions require confirmation and an optional/mandatory reason.
- Optimistic concurrency or row versioning shall prevent silent overwriting by two admins.

8.3 User and role management

- Super Admin can create, invite, activate, suspend and assign any application role.
- Manager can create and manage Regular Admin and Manager accounts, but cannot perform any action against a Super Admin identity or grant Super Admin.
- Regular Admin cannot access user management unless a narrowly scoped permission is explicitly introduced later.
- No role can delete its own final Super Admin account, and the system must prevent removal of the last active Super Admin.
- Account events are audited. Authentication credentials are owned by the identity provider, not stored by the application.

9. Data Model

Entity	Core fields / notes
User	id, identityProviderId, displayName, email, role, status, lastLoginAt, createdAt, updatedAt
Role/Permission	role, permission, scope, createdAt
Customer	id, name, email, phone, consent timestamps, communication preference
Address	id, customerId, lines, city, postalCode, geo fields, serviceAreaId, access notes

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Entity	Core fields / notes
Package	id, name, slug, descriptions, basePrice, pricingType, duration, inclusions, exclusions, active, displayOrder
AddOn	id, name, priceAdjustment, durationAdjustment, compatiblePackages, active
Booking	id, reference, customerId, addressId, packageId, requested slots, agreed slot, status, estimate, currency, duration, source, rowVersion
BookingRevision	id, bookingId, version, changeSet, changedBy, reason, createdAt
Agent	id, name, contact, active, skills, service areas, maxJobsPerDay
AvailabilityRule	id, agentId, recurrence/work hours, effective dates
TimeOff	id, agentId, type, start, end, status, notes
Assignment	id, bookingId, agentId, start, end, buffer, status
Message	id, bookingId, direction, channel, subject, body/template, providerMessageId, parentMessageId, deliveryStatus
Notification	id, recipient, type, bookingId, readAt, createdAt
AuditEvent	id, actor, action, entityType, entityId, before/after summary, IP/user agent subject to policy, timestamp
SystemSetting	key, encrypted/safe value reference, category, updatedBy, updatedAt

Data design requirements

Use UTC timestamps in storage and render in the configured business timezone. Use database constraints, foreign keys, soft delete where retention is required, indexes for reference/status/date searches, and an immutable audit pattern for critical events.

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10. API Requirements

Method	Route	Purpose
POST	/api/public/bookings	Create booking request; anonymous with anti-abuse controls.
GET	/api/public/packages	Return active packages and add-ons.
POST	/api/public/contact	Submit contact request.
GET	/api/admin/dashboard	Return authorized dashboard metrics.
GET/POST	/api/admin/bookings	Search/list or create bookings.
GET/PATCH	/api/admin/bookings/{id}	Read or update a booking with concurrency token.
POST	/api/admin/bookings/{id}/accept	Re-check capacity, assign agent(s), confirm and notify.
POST	/api/admin/bookings/{id}/propose-change	Create revision, notify customer and await response.
POST	/api/admin/bookings/{id}/reject	Reject with recorded reason and notification.
GET	/api/admin/availability	Return eligible agent availability for time range.
CRUD	/api/admin/agents	Manage agents and skills according to permission.
CRUD	/api/admin/packages	Manage package catalogue according to permission.
CRUD	/api/admin/users	Manage application-role mappings with hierarchy safeguards.
GET	/api/admin/audit	Search authorized audit events.
POST	/api/webhooks/email	Validate provider signature and process delivery/reply events.
GET/POST	/api/customer/bookings/{token}	Securely view/respond to a booking action.

- Use JSON over HTTPS, version APIs before breaking changes, and publish an OpenAPI specification.
- Return consistent error objects with correlation ID, safe user message, code and validation field details.
- Validate all inputs server-side. Ignore or reject client-supplied authorization fields.
- Apply authentication, permission checks, tenant restriction, rate limits, CSRF protections as applicable, idempotency and structured logging.
- Never trust availability calculated only in the browser.

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11. UX/UI and Responsive Design

11.1 Visual direction

The public experience should appear fresh, premium, clean and trustworthy rather than clinical or template-like. Use an AXA Cleaning design system with deep teal, aqua, bright white, subtle warm accents, generous whitespace, crisp typography, realistic service imagery or purpose-built 3D objects, and restrained glass-like surfaces. Animation must guide attention and demonstrate quality, not delay booking.

- Hero section with layered 3D cleaning-related art, strong headline, trust copy and Book Now/Compare Packages actions.
- Smooth section reveals, depth/parallax effects, high-quality hover states and page transitions that respect reduced-motion preferences.
- Interactive package cards and comparison drawer/modal with sticky selection actions.
- Clear booking stepper, autosaved in-session form state, inline validation and a persistent mobile call-to-action.
- Admin interface optimized for speed and density: accessible tables, calendar/timeline, drawers, bulk-safe actions and visible status tags.
- No critical information or action may depend on animation, hover, color alone, or a 3D-capable device.

11.2 Responsive breakpoints and accessibility

- Support at least 320 px mobile width through large desktop layouts.
- Meet WCAG 2.2 AA intent: semantic landmarks, keyboard access, logical focus, labels, contrast, error summaries, target size, captions/alt text and screen-reader announcements.
- Provide reduced-motion and low-performance fallbacks; defer or replace heavy 3D on constrained devices.
- Avoid horizontal page scrolling. Data tables may use controlled, labelled responsive patterns.
- Touch targets should be comfortable and forms must use appropriate mobile input types and autocomplete.

12. Security and Privacy

ID	Requirement
SEC-001	Use Microsoft Entra ID for staff authentication; restrict sign-in to the approved tenant and require MFA through organizational policy.
SEC-002	Enforce role and permission checks in every protected API. Client-side checks are usability only.
SEC-003	Do not store passwords. Do not seed real credentials in source, samples or configuration.
SEC-004	Use managed identities where possible; store remaining secrets in Key Vault and rotate them.
SEC-005	Encrypt data in transit and at rest using Azure platform capabilities and secure database configuration.
SEC-006	Apply validation, output encoding, parameterized queries/ORM safeguards, secure headers, dependency scanning and anti-abuse controls.
SEC-007	Log authentication, authorization failures, user management, booking changes, exports, overrides and settings changes.

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ID	Requirement
SEC-008	Separate customer-visible messages from internal notes and minimize personal information in logs.
SEC-009	Define retention, deletion/anonymization, subject-request and backup policies before production.
SEC-010	Use least-privilege production access, environment separation and protected GitHub branches with reviewed pull requests.
SEC-011	Threat-model IDOR, privilege escalation, booking enumeration, email spoofing/thread injection, duplicate submissions and race conditions.
SEC-012	Back up the database, test restore procedures and document incident response contacts and runbooks.

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13. Non-Functional Requirements

Quality attribute	Requirement
Performance	Target public LCP <= 2.5 seconds at the 75th percentile on representative mobile conditions; lazy-load 3D and noncritical media; keep interaction responsive.
Availability	Target 99.9% monthly availability for production application components, subject to selected Azure service tiers.
Scalability	Stateless API design and database indexes shall support growth without redesign; scheduled jobs must be idempotent.
Reliability	Booking creation and status updates must be transaction-safe. Email failures must not roll back a valid booking but must create a visible retry condition.
Security	No critical/high unresolved security findings at go-live; dependency and secret scanning enabled.
Accessibility	Critical customer and admin workflows pass keyboard-only and automated accessibility checks; manual screen-reader checks performed.
Compatibility	Latest two major versions of Edge, Chrome, Firefox and Safari; responsive iOS/Android browsers.
Maintainability	TypeScript strict mode, linting, formatting, modular architecture, tests, OpenAPI, migration scripts and documented environment variables.
Observability	Correlation IDs across UI/API/email; dashboards and alerts for error rate, latency, failed email, failed jobs and abnormal authorization failures.
Recovery	Define RPO/RTO with business owner; document backup, restore and redeployment procedures before launch.
Localization	All user-facing text externalized; currency, date/time and timezone configurable even if v1 launches in one locale.
SEO	Semantic HTML, metadata, canonical links, sitemap, structured data and optimized indexable public routes.

14. DevOps, GitHub and Azure Deployment

14.1 Repository structure

```
/
  src/                    # React public site and admin portal
  api/                    # Azure Functions API
  shared/                 # contracts, validation and shared types
  tests/                  # unit, integration, accessibility and E2E
  docs/                   # architecture, API, runbooks and decisions
  infra/                  # Bicep or Terraform
  public/                 # optimized static assets
  staticwebapp.config.json
```


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```
.github/workflows/
README.md
.env.example
```

14.2 Environments and pipeline

- Use development, staging/UAT and production environments with separate configuration and data stores.
- Pull requests run install, lint, type-check, unit tests, security checks, build and preview deployment.
- Main-branch merge deploys to staging or production according to release policy. Production should use protected environments/approvals.
- Infrastructure is defined as code. Manual portal changes are documented and then captured in code.
- Use database migrations with rollback planning. Seed only non-sensitive demonstration data outside production.
- Expose only public-safe build-time variables to the browser. Secrets belong in Azure configuration/Key Vault.
- Configure custom domain, HTTPS, DNS, email domain authentication, monitoring alerts and budget alerts before go-live.

15. Testing and Acceptance Criteria

15.1 Test coverage

- Unit tests for pricing, state transitions, permissions, availability, conflict checks and email correlation.
- Integration tests for database transactions, role enforcement, booking creation, assignment and notification records.
- End-to-end tests for customer booking and each administrative role.
- Accessibility, cross-browser, responsive, performance and reduced-motion testing.
- Security testing for unauthorized route/API access, IDOR, injection, XSS, CSRF where relevant, rate limits, secret exposure and privilege escalation.
- Operational testing for mail failure/retry, unavailable database, concurrent acceptance, backup/restore and redeployment.

15.2 Minimum acceptance scenarios

ID	Acceptance criterion
AC-01	A mobile customer compares packages, completes a valid request, receives a reference and acknowledgement, and the request appears in the admin queue.
AC-02	An admin sees eligible agents for the requested slot and can accept only after the server revalidates capacity.
AC-03	Two admins attempting to confirm conflicting work cannot silently double-book the same agent.
AC-04	A modified booking produces a revision, sends a customer update and retains the same booking conversation/reference.
AC-05	A Manager can manage operational users but cannot view or change a Super Admin account through UI or direct API.
AC-06	A Regular Admin cannot call package, user or protected settings endpoints without permission.
AC-07	A Super Admin can manage roles and settings, with every sensitive action audited.

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ID	Acceptance criterion
AC-08	Email delivery failure creates a visible retry/error item without losing the booking.
AC-09	All priority workflows work at 320 px width, by keyboard, and with reduced motion.
AC-10	A production deployment can be reproduced from GitHub and infrastructure/configuration documentation.

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16. Delivery Phases

Phase	Deliverables
0. Discovery	Confirm branding, packages, pricing, service areas, business hours, agent rules, email domain, privacy/retention, timezone and acceptance owners.
1. Foundation	Repository, design system, Azure resources, identity, infrastructure as code, CI/CD, database schema and observability.
2. Public experience	Commercial pages, package catalogue/comparison, responsive design, SEO and contact flow.
3. Booking	Multi-step booking, pricing estimate, confirmation, email acknowledgement and admin request queue.
4. Operations	Admin dashboard, availability/calendar, booking actions, agent schedules, role hierarchy and audit.
5. Hardening	Accessibility, security, performance, concurrency, retries, monitoring, backups and UAT.
6. Launch	Content migration, domain/email configuration, production validation, training, runbooks and hypercare.

17. Assumptions, Risks and Decisions

Item	Type	Treatment / decision
The brand name “AXA Cleaning” may overlap with another trademark.	Risk	Business owner must confirm legal right, domain and branding before launch.
Exact country, currency, taxes and service area are not yet supplied.	Assumption	Make them configuration-driven and confirm during discovery.
Package names/prices/content are not supplied.	Assumption	Use clearly marked demo content in non-production; require final content before UAT.
Email replies must remain in one conversation.	Technical risk	Use a transactional email provider with message IDs, reply handling/webhooks, booking reference and stored timeline.
3D effects can harm speed/accessibility.	UX risk	Use progressive enhancement, lazy loading, reduced-motion and lightweight fallback assets.
Azure Static Web Apps role mapping alone may not model every business permission.	Decision	Use identity-provider authentication plus API-side application RBAC and hierarchical safeguards.
Agent data may be personal information.	Privacy risk	Minimize public exposure, apply retention/access policy and audit access.

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Item	Type	Treatment / decision
Real-time notifications are desired but not mandatory for first release.	Decision	Start with in-app polling/refresh and email; optionally add Azure SignalR later.

18. Master Developer AI Prompt

You are the lead product engineer, solution architect, UX designer, security engineer, QA engineer and DevOps engineer for a production system named “AXA Cleaning”. Build the complete system described below. Do not produce a visual-only prototype. Deliver working front end, APIs, persistence, authentication, role authorization, notification records, tests, infrastructure, CI/CD and documentation.

PRIMARY OUTCOME
Create a premium commercial cleaning-service website and a secure administration portal. Host the front end on Azure Static Web Apps and maintain/deploy the solution from GitHub. Use a React + TypeScript + Vite front end and TypeScript Azure Functions APIs unless the existing repository has an equivalent approved stack. Use strict TypeScript, accessible components, responsive design and production-safe patterns.

NON-NEGOTIABLE SECURITY
1. Never store plaintext passwords and never build a custom password table. The public “Admin Portal” button must open secure Microsoft Entra ID authentication restricted to the approved tenant.
2. Enforce roles and permissions in every server API. UI hiding is not authorization.
3. Roles: Super Admin, Manager, Regular Admin. Super Admin has all business-application permissions. Manager has broad operational and user-management permissions but must never view, edit, disable, delete, reset, demote, assign roles to, or otherwise act upon a Super Admin account. Regular Admin can operate bookings and view agent availability but cannot manage protected users/settings unless explicitly granted a narrow permission.
4. Prevent removal of the last active Super Admin. Audit all security-sensitive actions.
5. Use Key Vault/managed identity or platform configuration for secrets. Never commit credentials, connection strings or production keys.

PUBLIC EXPERIENCE
Build Home, Services, Packages, Compare, About, Areas Served, Reviews, FAQ, Contact, Privacy, Terms and Book pages. The design must feel premium, clean, trustworthy and eye-catching, with deep teal/aqua/white design tokens, strong typography, purpose-built 3D-inspired cleaning visuals, polished micro-interactions and advanced but restrained animation. Support prefers-reduced-motion and low-performance fallbacks. Do not allow visual effects to block content, accessibility or Core Web Vitals.

PACKAGE AND BOOKING EXPERIENCE
- Configurable packages: names, descriptions, from/fixed price, estimated duration, size limits, inclusions, exclusions, badge, media, recommended frequency, display order and active state.
- Configurable add-ons with price/duration changes and compatible packages.
- Compare two to four packages side by side.
- Mobile-first multi-step booking: package/add-ons; property/address; customer/contact; preferred slot and alternatives; special requirements; review/consents; confirmation.
- Clearly state that the selected time is requested, not confirmed, until accepted by AXA Cleaning.
- Generate a human-readable booking reference and idempotently create the request.
- Send acknowledgement email and show confirmation. Keep customer-visible messages separate from internal notes.

ADMIN PORTAL
Build a responsive protected portal with:
- Dashboard KPIs, new request queue, overdue actions, email failures and cleaning-agent availability.
- Day/week/month calendar or timeline of agent availability and assignments.
- Booking list, kanban and calendar views; powerful search/filter; booking detail timeline.
- Booking actions: create, accept, propose modification, assign/reassign agents, reject, cancel, start, complete, no-show, resend notification and export/print summary.
- Agent management: active status, contacts, skills, service areas, recurring schedules, exceptions, leave, buffers and maximum jobs/day.
- Package/add-on/content management according to role.
- User/role management with the hierarchy safeguards above.
- Audit viewer, notification center, reports and system settings.

BOOKING WORKFLOW
Implement statuses: Draft, Requested, Under Review, Awaiting Customer, Accepted, Modified, In Progress, Completed, Rejected,

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Cancelled, No-show and Closed. Validate allowed transitions server-side. Record actor, timestamp, reason and revision. Re-check availability in the same transaction that confirms a booking. Prevent overlapping assignments including travel/setup buffers. Use concurrency control so two admins cannot silently overwrite or double-book.

EMAIL AND NOTIFICATIONS
Use Azure Communication Services Email or a provider abstraction. Send acknowledgement, acceptance, modification proposal, rejection, cancellation, reminder and completion emails. Preserve a stable booking reference and conversation identifier. Store provider message IDs, delivery events and the customer-visible message timeline. Where supported, use Message-ID, In-Reply-To and References so updates remain in the same mail conversation. Provide secure inbound reply correlation/webhook validation or a documented adapter. Retry transient failures and expose persistent failures in the admin action queue. Company sender name, address, logo, footer and support details must be configurable.

DATA AND API
Use Azure SQL Database by default with migrations, constraints, indexes, timestamps in UTC, soft deletion/retention behavior and row-version concurrency. Create entities for users/roles/permissions, customers, addresses, packages, add-ons, bookings, revisions, agents, schedules, time off, assignments, messages, notifications, audit events and settings. Implement REST APIs under /api with structured errors, correlation IDs, validation, rate limiting, idempotency and an OpenAPI document. Protect every admin endpoint. Do not trust price, role, customer identity, status transition or availability values supplied by the browser.

REPOSITORY AND AZURE
Use a clear monorepo structure: src, api, shared, tests, docs, infra and .github/workflows. Include staticwebapp.config.json. Provide Bicep or Terraform for Azure resources. Configure preview environments for pull requests and protected deployment to dev/staging/production. Pipelines must run install, formatting/lint, type-check, unit tests, integration tests, accessibility checks, security/dependency/secret scans and production build. Add Application Insights telemetry, correlation IDs, alerts and runbooks. Include .env.example with placeholders only.

QUALITY GATES
- Responsive from 320 px to large desktop; no page-level horizontal scrolling.
- WCAG 2.2 AA intent and keyboard-complete critical workflows.
- Public LCP target <= 2.5 seconds at p75 on representative mobile conditions; lazy-load 3D/media.
- Latest two major versions of Edge, Chrome, Firefox and Safari.
- Unit tests for permissions, state transitions, pricing, availability and email correlation.
- Integration/E2E tests for customer request, each admin role, concurrent booking acceptance, notification failure and protected endpoints.
- No unresolved critical/high security issue at release.

IMPLEMENTATION METHOD
1. Inspect the repository and write docs/implementation-plan.md mapping requirements to modules and milestones.
2. List assumptions and unresolved business inputs, but do not stop. Use configuration-driven placeholders and clearly label demo content.
3. Create architecture, data model, API contract, route map and role-permission matrix before completing UI modules.
4. Build vertical slices: foundation/auth; public packages; booking request; admin queue; availability/acceptance; notifications; management/reporting; hardening.
5. After each slice, run tests and fix failures. Do not claim completion based only on generated code.
6. Provide setup scripts, Azure deployment guide, local development instructions, seed/demo data, test accounts via identity-role mapping instructions (never passwords), operations runbook and UAT checklist.
7. Maintain a requirement traceability checklist in docs/requirements-traceability.md. Mark an item complete only when implementation and tests exist.

DEFINITION OF DONE
The project is done only when it builds successfully, automated tests pass, protected APIs reject unauthorized/incorrect roles, the customer can submit a booking, an admin can safely process it against real stored availability, updates are persisted/audited/notified, the design is responsive/accessibility-checked, and the solution can be reproducibly deployed from GitHub to Azure using the supplied infrastructure and documentation.

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Appendix A - Discovery Inputs Required

- Official logo, brand colors, approved legal name and domain.
- Launch country/currency, tax/fee rules, timezone, languages and service areas.
- Package catalogue, pricing formula, minimum charge, add-ons and estimated durations.
- Business hours, booking notice period, cancellation policy, slot length and travel buffers.
- Agent list, skills, work patterns, leave approval process and capacity rules.
- Official sender email, support email/telephone, reply-handling requirement and email-domain DNS access.
- Privacy/retention policy, terms, cookie/analytics decision and feedback/review source.
- Azure tenant/subscription, GitHub organization/repository and production approval owners.

Appendix B - References

Microsoft Learn. “Authenticate and authorize Static Web Apps.” Azure Static Web Apps documentation. https://learn.microsoft.com/azure/static-web-apps/authentication-authorization

Microsoft Learn. “Assign Azure Static Web Apps roles with Microsoft Graph (preview).” https://learn.microsoft.com/azure/static-web-apps/assign-roles-microsoft-graph

Note: The implementation team must validate current Azure features, service limits, pricing, regional availability and licensing at build time.

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